

VISION AND MISSION OF THE DEPARTMENT

VISION: Our vision is to develop the department into a full-fledged center of learning in various fields of Electronics & Communication Engineering keeping in view the latest developments.

MISSION: The Mission of the department is to turn out full-fledged Engineers in the field of Electronics Communication Engineering with an overall background suitable for making a successful career either in industry/research or higher education in India and abroad. To inculcate professional behavior, strong ethical values, innovative research capabilities and leadership abilities in the young minds so as to work with a commitment to the progress of the nation.

PROGRAM EDUCATIONAL OBJECTIVES

PEO 1: Excel in professional career & higher education, by acquiring knowledge in related fields of Electronics & Communication Engineering.

PEO 2: Exhibit leadership in their profession, through technological ability and contemporary knowledge for solving the real life problems appropriately that are technically sound, economically feasible & socially acceptable.

PEO 3: Adapt to the emerging technologies for sustenance by exhibiting professionalism, ethical attitude & communication skills in their relevant areas of interest by engaging in lifelong learning.

PROGRAM SPECIFIC OUTCOMES

A graduate of the Electronics and Communication Engineering Program will demonstrate:

PSO 1: System Design and Development: Ability to design and develop a secure electronic voting system by integrating fingerprint sensor, keypad, microcontroller, and display modules for reliable voting operations.

PSO 2: Hardware and Software Implementation: Ability to apply embedded system concepts, programming skills, and modern tools for implementing real-time authentication, vote recording, and duplicate vote prevention.

PSO 3: Innovation and Security Applications: Ability to develop innovative and user-friendly secure voting solutions using biometric technology to improve transparency, efficiency, and trust in election systems.

PROGRAM OUTCOMES

PO 1: Engineering Knowledge: An ability to apply knowledge of mathematics, science, fundamentals of engineering to solve electronics and communication engineering problems.

PO 2: Problem Analysis: An ability to identify, formulate and analyze and solve complex electronics and communication Engineering using the first principles of mathematics and engineering sciences.

PO 3: Design/Development of Solutions: An ability to develop solutions to electronics and communication systems to meet the specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.

PO 4: Conduct Investigation of Complex Problems: An ability to design and perform experiments of electronic circuits and systems, analyze and interpret data to provide valid conclusions.

PO 5: Modern Tool Usage: An ability to learn, select and apply appropriate techniques, resources and modern engineering tools including prediction and modeling, to complex electronics and communication systems.

PO 6: The Engineer and Society: An ability to assess the knowledge of contemporary issues to the societal responsibilities relevant to the professional practice.

PO 7: Environment and Sustainability: An ability to understand the impact of professional engineering solutions in societal and environmental contexts and demonstrate knowledge for the need of sustainable development.

PO 8: Ethics: An ability to demonstrate the understanding of professional, ethical responsibilities and norms of engineering practice.

PO 9: Individual and Team Work: An ability to function effectively as an individual and as a member or leader in diverse teams and in multidisciplinary settings.

PO 10: Communication: An ability to communicate effectively with engineering community and with society at large.

PO 11: Project Management and Finance: An ability to demonstrate knowledge and understanding of engineering and management principles and apply these to manage projects.

PO 12: Life Long Learning: An ability to recognize the need for, and engage in lifelong learning in the broadest context of technological change.

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ACRONYMS

Acronym	Meaning
EVS	Electronic Voting System
UID	Unique Identification
OTP	One Time Password
LCD	Liquid Crystal Display
MCU	Microcontroller Unit
GPIO	General Purpose Input Output
ADC	Analog to Digital Converter
DC	Direct Current
AC	Alternating Current
RFID	Radio Frequency Identification

ABSTRACT

This project presents a **Biometric Electronic Voting System with Real-Time Authentication using a Keypad**, designed to improve security, transparency, and reliability in voting processes.

Traditional voting systems are vulnerable to fraud, impersonation, and manual errors. To overcome these challenges, this system integrates **biometric fingerprint authentication along with keypad-based verification**, ensuring that only authorized voters can cast their votes.

The system uses a fingerprint sensor to identify voters uniquely, while the keypad allows secure input of voter ID or verification code. An Arduino-based microcontroller processes the inputs and enables voting only after successful authentication.

This approach eliminates duplicate voting, enhances accuracy, and ensures real-time validation. The system is user-friendly, cost-effective, and suitable for deployment in secure electronic voting applications.

1. INTRODUCTION

1.1 INTRODUCTION

Electronic Voting System (EVS), are modern solution designed to replace traditional paper – based voting methods with secure, efficient, and reliable digital systems. Conventional voting methods often suffer from issue such as vote duplication, impersonation, manual counting errors, and lack of transparency.

To overcome these limitations, this project introduces an **Electronic Voting System with Real Time Authentication and Keypad**, which ensure that only authorized individuals can cast their votes.

1.2 INTRODUCTION TO ELECTRONIC VOTING SYSTEM:

An Electronic Voting System is a technology-based system that allows voters to cast their votes electronically instead of using traditional ballot papers.

In a typical electronic voting system:

- Votes are recorded digitally
- Counted is automated
- Results are generated quickly and accurately

Advantages of Electronic Voting Systems:

- Faster vote counting
- Reduced human errors
- Improved transparency
- Easy data storage and retrieval

1.3 INTRODUCTION TO BIOMETRIC AUTHENTICATION:

Biometric authentication is a security process that verifies a person's identity based on unique biological characteristics such as:

- Fingerprints
- Iris Patterns
- Facial recognition

In this project, **fingerprint recognition** is used as the primary authentication method because:

- It is unique for every Individual

- It is difficult to duplicate
- It provides high Accuracy

The fingerprint sensor captures and compares the user's fingerprint with stored data. If a match is found, access is granted to proceed with your voting is counted.

1.4 APPLICATIONS OF BIOMETRIC ELECTRONIC VOTING:

Biometric voting systems have a wide range of applications in various sectors:

- **Government Elections**
Ensures fair and transparent voting by preventing duplicate votes.
- **College and University Elections**
Used for student elections to maintain accuracy and eliminate malpractice.
- **Corporate Voting Systems**
Helps organizations conduct secure decision-making processes.
- **Local Governance (Panchayat / Municipal Voting)**
Provides a cost-effective and secure voting solution.
- **Secure Access Control Systems**
Can be adapted for attendance and identity verification systems.

1.5 TYPES OF AUTHENTICATION METHODS:

Authentication is a crucial part of any secure system. Various authentication methods are used to verify user identity:

1.5.1: Fingerprint Authentication:

- Uses unique Fingerprint patterns
- Highly secure and widely used
- Difficult to replicate

1.5.2 Keypad – Based Authentication:

- Requires input of PIN or voter ID
- Acts as a secondary security layer
- Easy to implement and user-friendly

1.5.3 OTP-Based Authentication

- One Time Password sent to registered mobile number
- Provides real-time verification
- Prevents unauthorized access

1.5.4 Multi-Factor Authentication

- Combines multiple authentication methods
- Example: Fingerprint + Keypad
- Provides maximum security

2 LITERATURE REVIEW

Biometric Electronic Voting Systems have gained significant attention in recent years due to the increasing need for secure, transparent, and efficient election processes. Various researchers have proposed different techniques to improve the reliability and security of electronic voting system.

Early electronic voting systems primarily relied on simple digital interfaces without strong authentication mechanisms. According to Kohno et al. (2004), traditional electronic voting machines were vulnerable to security threats such as tampering, unauthorized access, and malware attacks. This highlighted the necessity for stronger voter authentication methods.

To address this issue, researchers introduced biometric – based authentication systems. Jain et al. (2011) emphasized that biometric traits like fingerprints provide a unique and reliable method for identifying individuals, making them suitable for secure voting applications. Their work demonstrated that biometric systems significantly reduce the risk of impersonation.

Further advancements were made by integrating fingerprint recognition with microcontroller – based systems. Rath et al. (2013) developed a fingerprint – based voting system using embedded systems, which improved authentication accuracy and reduced manual intervention.

However, relying solely on biometric authentication was found to have limitations, such as sensor errors and spoofing risks. To overcome these, researchers proposed multi – factor authentication systems. Patel and Shah (2016) introduced a system combining biometric verification with PIN – based authentication, enhancing overall system security.

In recent studies, real – time authentication mechanisms have been incorporated into voting systems. Singh et al. (2019) proposed a real – time biometric voting system that validates voter identity instantly and prevents duplicate voting. Their system demonstrated improved efficiency and faster processing.

3 EXISTING METHOD

- ❖ The traditional electronic voting systems mainly rely on manual or basic digital authentication, such as voter ID cards or simple login credentials.
- ❖ Most systems use single – factor authentication, which is not highly secure and can allow unauthorized access.
- ❖ Paper – based voting systems are still widely used which involve manual counting, leading to errors and delays.
- ❖ Existing electronic voting machines (EVMs) do not always include biometric verification, making them vulnerable to impersonation and duplicate voting.

4 PROPOSED METHOD

- A proposed an Electronic Voting System with Real Time Authentication and Keypad, designed to enhance security, accuracy and transparency in voting.
- The systems use fingerprint authentication as the primary method to verify the identity of the voter. Each voter must enroll their fingerprint in the system database prior to voting.
- In addition to biometric verification, a based authentication (such as entering a voter ID or PIN) is used as a secondary layer of security. This ensures multi – factor authentication.

5 DESIGN CONSIDERATIONS

5.1 BLOCK DIAGRAM:

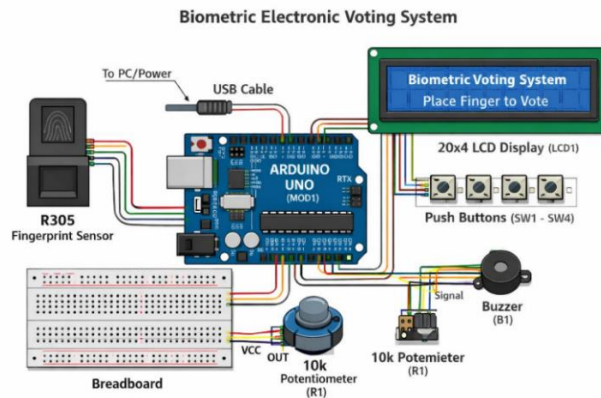


Figure 5.1: Block Diagram

The block diagram consists of the Arduino UNO, Fingerprint sensor (R05), LCD Display, push buttons, buzzer, and power supply as shown in figure 1. Arduino acts as the main controller which manages all the operations of the system

After successful authentication, the Arduino the voting process. The LCD display shows instructions and status messages, while push buttons are used by voter to select the candidate.

5.2: SYSTEM ARCHITECTURE:

System Architecture of the Biometric Electronic Voting system consists of the integrated structure where the fingerprint sensor, Arduino UNO, LCD display, push buttons and buzzer work together to secure voting operations. The fingerprint sensor captures verifies the user's identity and Arduino process this data to allow or restrict access to the voting system operation as shown in figure 2.

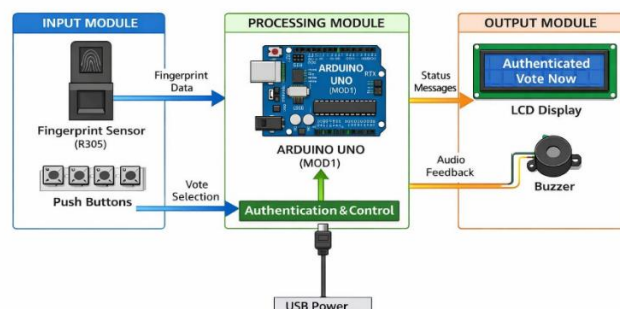


Figure 5.2: System Architecture

Feature	Traditional Voting	Electronic Voting (EVM)	Biometric Voting System
Method	Paper ballot	Electronic machine	Fingerprint-based system
Authentication	Voter ID/manual check	Voter ID/manual check	Biometric (fingerprint)
Security Level	Low	Medium	High
Duplicate Voting	Possible	Limited	Not possible
Speed	Slow	Faster	Very Fast
Accuracy	Less accurate	Accurate	Highly accurate
Fraud Chances	High	Medium	Very Low
Cost	Low, high maintenance	Moderate	Higher initial cost

Table 5.1: Difference between Voting System

5.3: POWER SUPPLY:

The power supply unit will provide the required +5V DC voltage for the operation of the Biometric Electronic voting System. In the project, a USB cable is used as the power source, which is connected to the Arduino UNO. The USB interface supplies a regulated 5V from a computer or adapter.

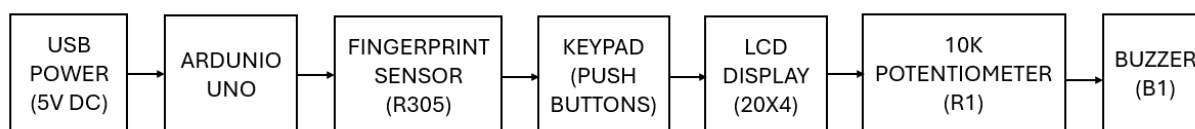


Figure 5.3: Block diagram of Power Supply

5.3.1: PUSH TO ON SWITCH (SW1 – SW2):

Push to ON switches such as SW1 and SW2 are momentary switches that remain on OFF state normally and become ON only when pressed. In this system, they are used to select candidate after successful authentication. Once the buttons are released, it returns to its original OFF state, to preventing multiple votes from a single press.

These switches are connected using pull - down or pull - up resistors to ensure stable inputs signals. They play crucial role in enabling controlled and accurate voting in the system.



Figure 5.4: Push to ON Switch (SW1-SW2)

5.3.2: ARDUINO UNO:

The Arduino UNO is the main microcontroller board used in this Biometric Electronic Voting system. It is based on the ATmega328p microcontroller and is responsible for an act as the central processing unit that controls and coordinates all systems input and output operations such as fingerprint authentication, vote recording, display output, buzzer indication.

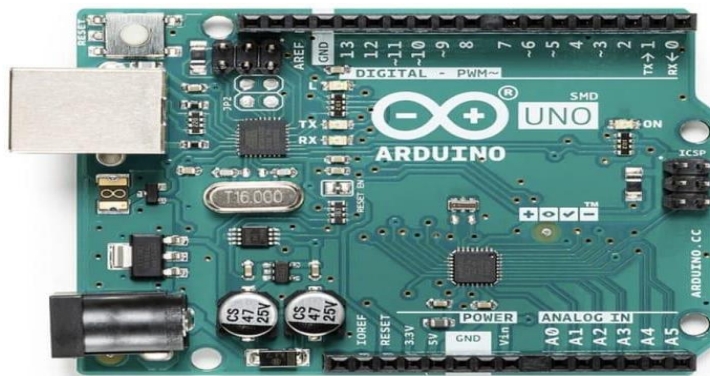


Figure 5.5: Arduino UNO

It has input/output, 6 analog input pins, a USB Connection, and operates at 5V. The Arduino receives input, from the fingerprint sensor and push buttons, processes the data and controls the output devices like the LCD display and buzzer etc.

Working In this project:

In the voting system, Arduino UNO performs the following functions.

- Receivers Fingerprint data from R305 sensor
- Verifies whether the voter is authorized
- Enables voting only after successful authentication
- Reads input from push buttons (keypad) for candidates' selection
- Display's message on the 20x4 LCD
- Activates buzzer for alerts (success/error)

Why Arduino UNO is used:

- Easy to program and user – friendly
- Supports multiple devices at the same time
- Low – cost and widely

5.3.3: FINGERPRINT SENSOR (R305):

The Fingerprint sensor (R305) is a biometric device used to identify and verify a person based on their fingerprint. Every person has a unique fingerprint, so it is used for secure authentication in the voting system.



Figure 5.6 Fingerprint Sensor (R305)

Why it is used:

- Prevents fake voting
- Ensures one person to one vote
- If not matched to access is denied

5.3.4: 20X4 LCD DISPLAY (LIQUID CRYSTAL DISPLAY):

The 20x4 LCD (Liquid Crystal display) is used to show message and instructions Biometric Electronic voting System. It can display 20 Characters in each row and 4 rows, making suitable for clearly presenting information to the user. In this project, the LCD displays message such as “Place Finger”, “Access Granted”, “Select Candidate”, and “Vote Casted”. It is connected to the Arduino UNO, which sends data to be displayed based on the system operation.



Figure 5.7: LCD (LIQUID CRYSTAL DISPLAY)

5.8: 10K POTENTIOMETER:

The 10k potentiometer is a variable resistor used to control voltage in the circuit. It has three terminals: VCC, GND, and Output (middle pin). By rotating the knob, the resistance changes, which adjusts the output voltage.

In this Biometric Electronic Voting System, the potentiometer is mainly used to adjust the contrast of the 20×4 LCD display. By turning the knob, the text on the LCD becomes clearer or dimmer, improving readability.



Figure 5.11: 10k potentiometer

5.9: MICROCONTROLLER:

The microcontroller is the main processing unit of an embedded system, often called the brain of the system. It is a small integrated circuit that can read inputs, process data, and control outputs based on programmed instructions.

In the Biometric Electronic Voting System, the microcontroller (ATmega328P in Arduino Uno) controls all operations. It receives input from the fingerprint sensor and push buttons, processes the data, and sends signals to the LCD display and buzzer. The microcontroller authenticated users can vote, manages voting process, stores the voting data.

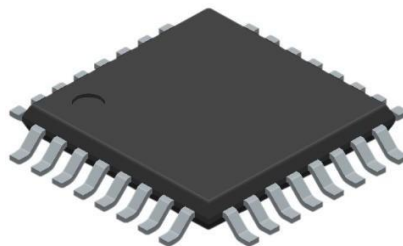


Figure 5.11: Microcontroller

5.10: ATMEGA328 MICROCONTROLLER:

The ATmega328P is an 8-bit microcontroller used in the Arduino Uno board. It is the main

processing unit that executes the program and controls all operations in the system. It has 32KB Flash memory, 2KB SRAM, and 1KB EEPROM, which are used to store code and data.

In the Biometric Electronic Voting System, the ATmega328P reads input signals from the fingerprint sensor and push buttons, processes the authentication, and controls output devices like the LCD display and buzzer.

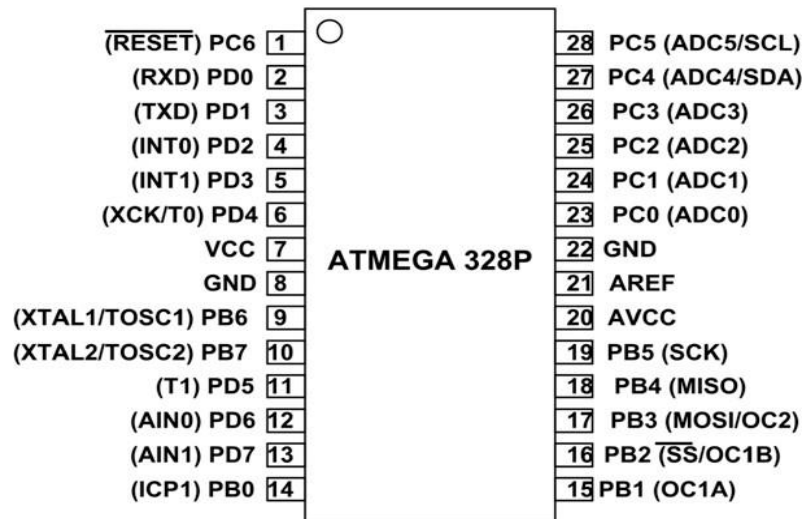


Fig.5.12: Pin diagram of Atmega328

5.11: PIN DESCRIPTION:

The ATmega328P microcontroller has 28 pins, which are used for power supply, input/output operations, and communication.

1. Power Pins

- **VCC** → Supplies +5V power to the microcontroller
- **GND** → Ground connection
- **AVCC** → Power supply for analog section
- **AREF** → Reference voltage for analog inputs

2. Digital I/O Pins

- **PORTB (PB0–PB5)** → Digital pins (used for input/output, PWM, SPI)
- **PORTD (PD0–PD7)** → Digital pins (used for general I/O, serial communication)

These pins are used in the project to connect:

- Push buttons
- Buzzer
- LCD

3. Analog Input Pins

- **PORTC (PC0–PC5)** → Analog input pins (A0–A5)

Used to read analog signals if needed.

4. Serial Communication Pins

- **PD0 (RX)** → Receive data
- **PD1 (TX)** → Transmit data

Used for communication with:

- Fingerprint sensor
- Computer (USB)

5. Reset Pin

- **RESET (PC6)** → Restarts the microcontroller when triggered

6. Crystal Oscillator Pins

- **XTAL1 & XTAL2** → Used for clock generation (16 MHz).

5 SOFTWARE DEVELOPMENT

Arduino Software:

It is necessary to download the Arduino Software package for your operating system. When you have downloaded and opened the application you should see something like this:

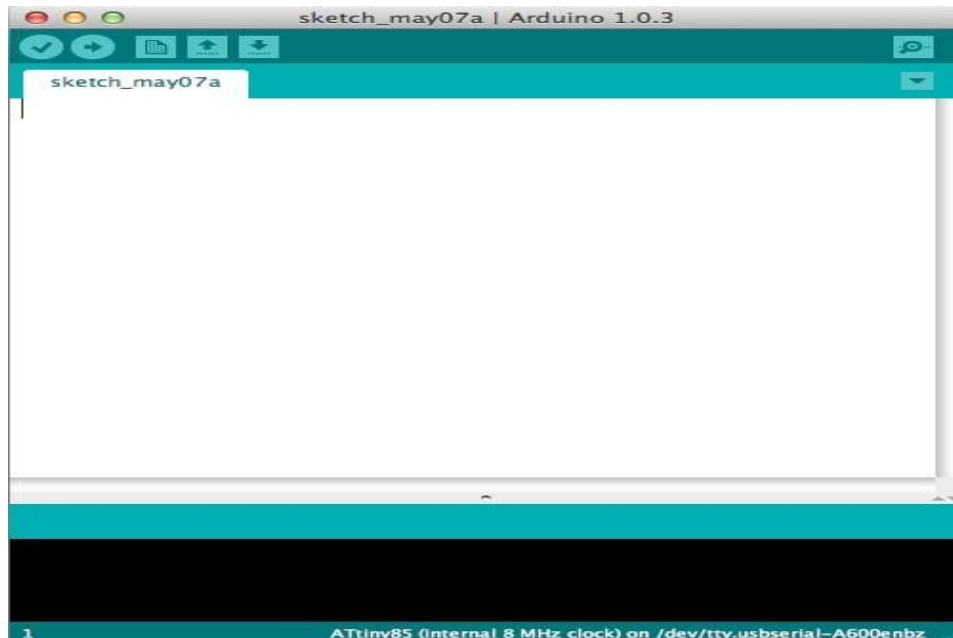


Figure 6.1: Arduino IDE

This is where you type the code you want to compile and send to the Arduino board.

The code you write for your Nodemcu esp8266 are known as sketches. They are written in C++.

Every sketch needs two void type functions, `setup ()` and `loop ()`. A void type function does not return any value.

The `setup ()` method is run once at the just after the ESP8266 is powered up and the `loop ()` method is run continuously afterwards. The `setup ()` is where you want to do any initialization steps, and in `loop ()` you want to run the code you want to run over and over again.

The Initial Setup:

We need to setup the environment to Tools menu and select Board.

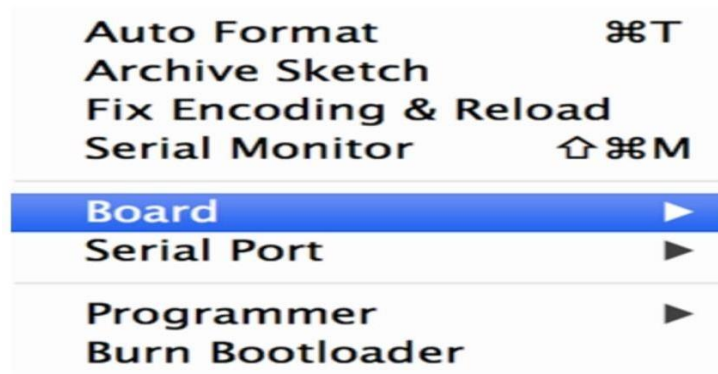


Figure 6.2: Select Board

Then select the type of Arduino you want to program in our case it is the ESP8266. The code you write for your Nodemcu esp8266 are known as sketches. They are written in C++.

Compiling the Code

If this is your first time you have ever compiled code to your Arduino before plugging it in to the computer go to the Tools menu, then Serial Port and take note of what appears there. like before plugging in the Arduino UNO:

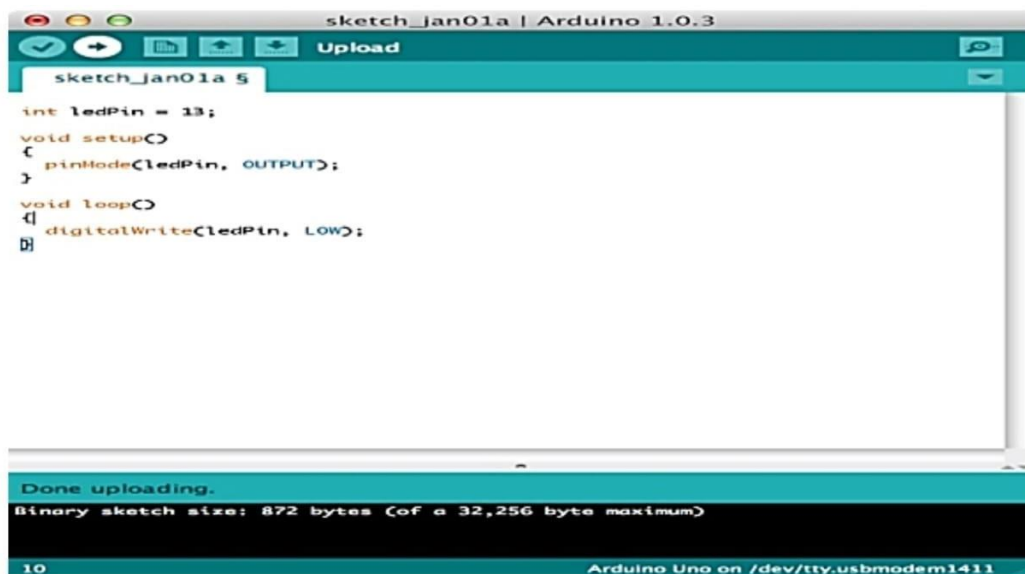


Figure 6.3: Compiling Uploading Program

Plug your Arduino UNO board in to the USB cable and into your computer. Now go back to the Tools > Serial Port menu and you should see at least 1 new option of serial ports appear.

- A USB A-to-B cable shown in fig 6.3, or another appropriate way to connect your Arduino-compatible microcontroller to your computer (check out this USB buying guide if you're not sure which cable to get).

On Windows you should see COM followed by a number. Select the new one that appears. Once you have selected your serial or COM port you can then press the button with the arrow pointing to the right.

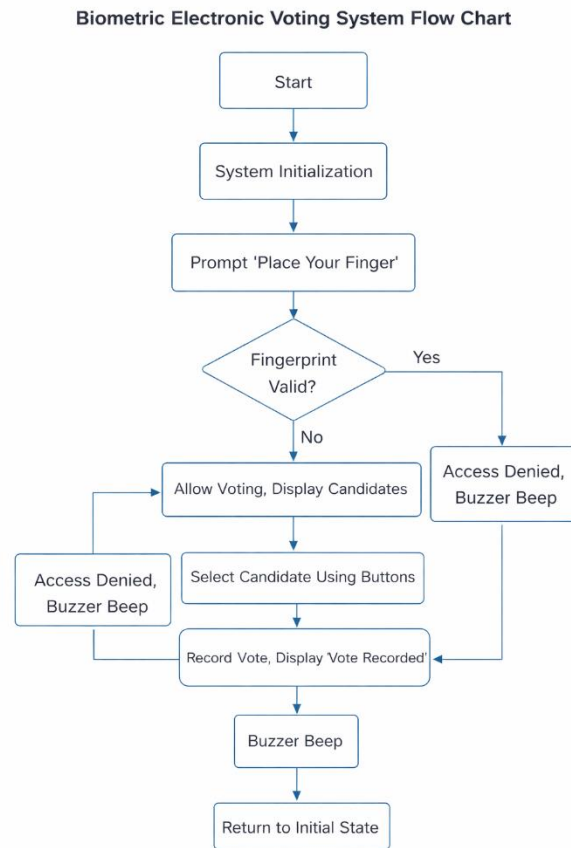


Figure 6.5: Flow Chart

5 IMPLEMENTATION

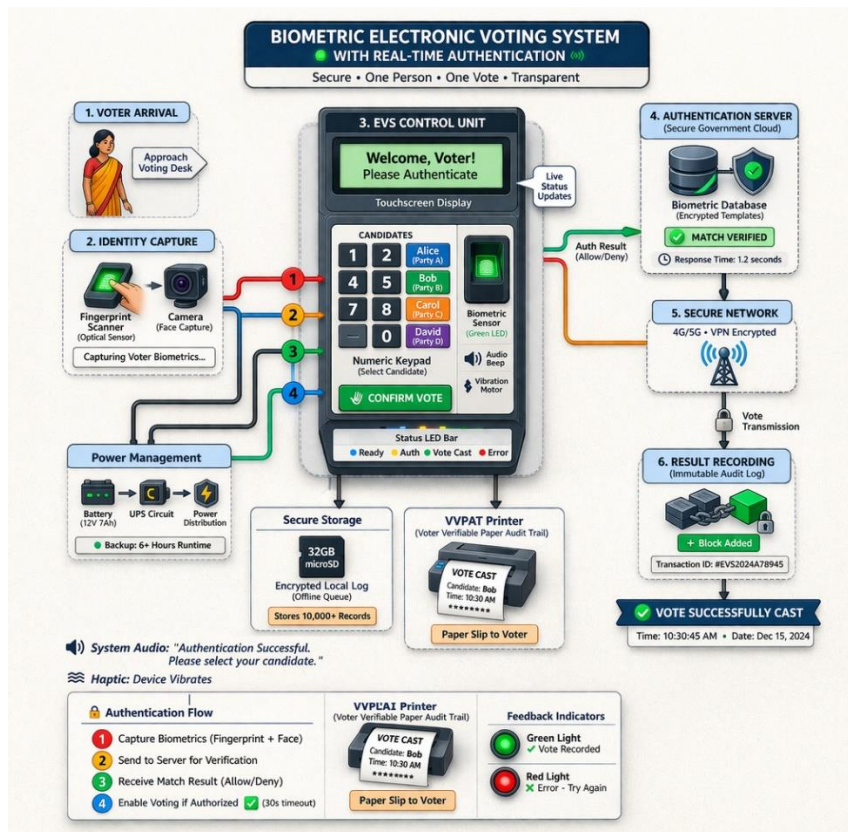


Fig.7.1: Schematic Diagram

The implementation of the Biometric Electronic Voting System with Real-Time Authentication is carried out by integrating both hardware and software components. All the components such as Arduino Uno, fingerprint sensor, LCD display, push buttons, buzzer, and potentiometer are connected properly using a breadboard and jumper wires.

Initially, the fingerprint sensor is interfaced with the Arduino using serial communication (TX and RX pins). The LCD display is connected to display instructions and voting status. Push buttons are connected to digital pins and act as voting inputs, while the buzzer is connected to provide audio feedback.

The system is programmed using Arduino IDE. The program includes fingerprint enrolment and matching, button detection, vote recording, and display control. Once powered through the USB cable, the system starts by asking the user to place their finger. If the fingerprint matches with stored data, the system allows the user to vote by pressing a button.

After voting, the system records the vote and displays a confirmation message on the LCD. If an unauthorized user attempts to vote, access is denied and the buzzer alerts the user.

5.1 ADVANTAGES:

- High security using fingerprint authentication
- Prevents duplicate and fake voting
- Easy to use and user-friendly
- Fast and real-time operation
- Reduces human errors

5.2 APPLICATIONS:

- College and university elections
- Organization and office voting systems
- Small-scale government elections
- Secure access control systems
- Survey and polling systems

6 RESULT ANALYSIS

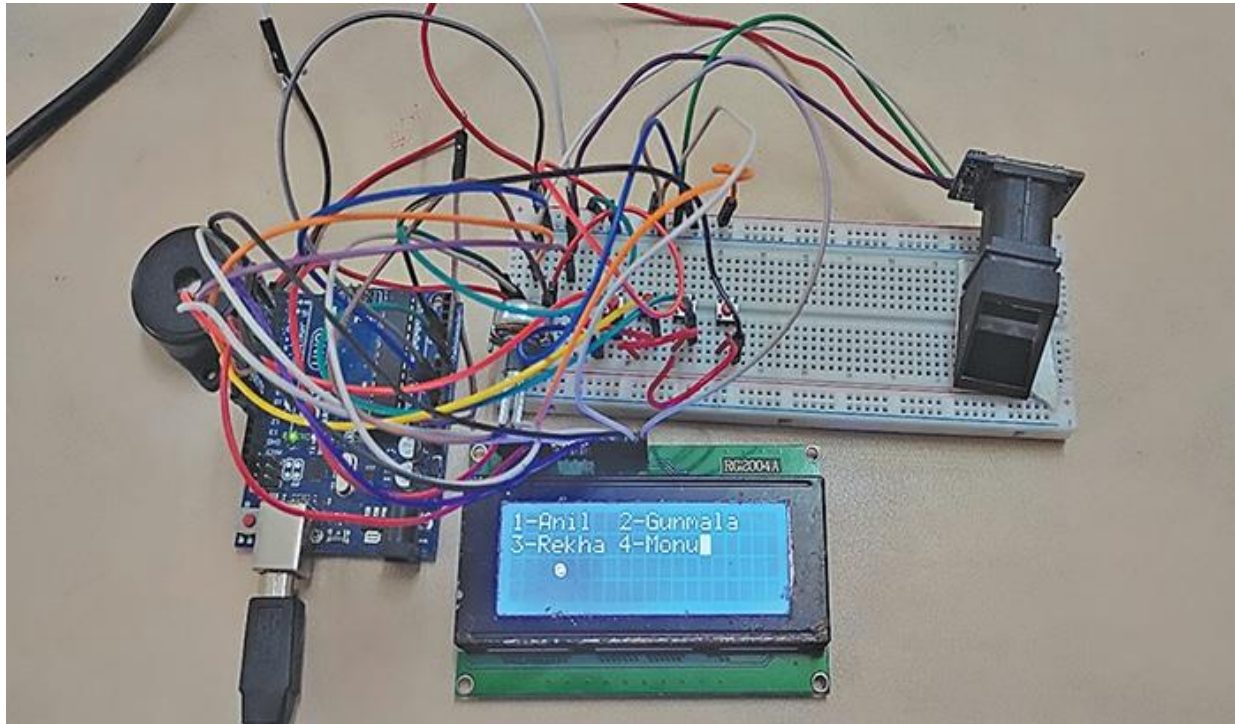


Figure 8.1: complete project photo

The Biometric Electronic Voting System was tested under different conditions to evaluate its performance. The fingerprint sensor successfully identified registered users and denied access to unregistered users, ensuring secure authentication. The system responded quickly, providing real-time feedback through the LCD display and buzzer.

During testing, each voter was allowed to vote only once, and duplicate voting was prevented effectively. The push buttons worked accurately for candidate selection, and the votes were recorded correctly without any data loss. The LCD displayed clear messages at every stage, improving user interaction.

Overall, the system showed reliable performance with high accuracy in authentication and voting. The results confirm that the system is efficient, secure, and suitable for small-scale voting applications.

7 CONCLUSION

The Biometric Electronic Voting System with Real-Time Authentication using keypad provides a secure, reliable, and efficient solution for modern voting requirements. By integrating fingerprint-based authentication, the system ensures that only authorized users can cast their votes, thereby eliminating issues like duplicate voting, impersonation, and manual errors.

The use of Arduino Uno as the controller makes the system simple, cost-effective, and easy to implement. The fingerprint sensor enhances security, while the LCD display and buzzer improve user interaction by providing clear instructions and feedback. The push-button keypad allows easy candidate selection, making the system user-friendly for all types of users.

The implementation and testing of the system showed accurate results with quick response time and stable performance. The system successfully handled authentication, voting, and result recording without errors. It can be effectively used in colleges, organizations, and small-scale elections where security and transparency are important.

In conclusion, this project demonstrates that biometric technology can significantly improve the voting process by making it more secure, transparent, and efficient. With further enhancements, the system can be expanded for large-scale elections and integrated with advanced technologies for better performance.

8 FUTURESCOPE

- Integration with **cloud database** for storing voter data and results securely
- Use of **IoT technology** for remote monitoring and centralized control
- Implementation of **face recognition or iris scanning** for multi-level security
- Development of **mobile or web-based voting interface**
- Adding **wireless communication (Wi-Fi/Bluetooth)** for data transmission

9 REFERENCES

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- [9] Online Resources: www.electronicshub.org, www.circuitdigest.com
- [10] Arduino IDE Documentation and Tutorials

APPENDIX

```
#include <Adafruit_Fingerprint.h>

#include <SoftwareSerial.h>

#include <LiquidCrystal.h>

#include <Keypad.h>


// Fingerprint setup

SoftwareSerial mySerial(2, 3);

Adafruit_Fingerprint finger = Adafruit_Fingerprint(&mySerial);


// LCD setup (RS, E, D4, D5, D6, D7)

LiquidCrystal lcd(7, 6, 5, 4, 3, 2);


// Keypad setup

const byte ROWS = 4;

const byte COLS = 4;


char keys[ROWS][COLS] = {

  {'1','2','3','A'},

  {'4','5','6','B'},

  {'7','8','9','C'},

  {'*','0','#','D'}

};


byte rowPins[ROWS] = {A0, A1, A2, A3};

byte colPins[COLS] = {A4, A5, A6, A7};


Keypad keypad = Keypad(makeKeymap(keys), rowPins, colPins, ROWS, COLS);
```

```
// Vote storage

int candidate1 = 0;
int candidate2 = 0;
int candidate3 = 0;

bool voted[50]; // store voted users (ID based)

void setup() {
    Serial.begin(9600);
    finger.begin(57600);

    lcd.begin(16, 2);
    lcd.print("Voting System");
    delay(2000);
    lcd.clear();

    if (finger.verifyPassword()) {
        lcd.print("Sensor Ready");
    } else {
        lcd.print("Sensor Error");
        while (1);
    }
    delay(2000);
    lcd.clear();
}

void loop() {
```

```
lcd.setCursor(0, 0);  
lcd.print("Place Finger");  
  
int id = getFingerprintID();  
  
if (id != -1) {  
    if (voted[id] == true) {  
        lcd.clear();  
        lcd.print("Already Voted");  
        delay(2000);  
        lcd.clear();  
        return;  
    }  
  
    voted[id] = true;  
  
    lcd.clear();  
    lcd.print("1:A 2:B 3:C");  
  
    while (true) {  
        char key = keypad.getKey();  
  
        if (key) {  
            if (key == '1') {  
                candidate1++;  
                break;  
            } else if (key == '2') {  
                candidate2++;  
            }  
        }  
    }  
}
```

```
        break;
    } else if (key == '3') {
        candidate3++;
        break;
    }
}

}

}

lcd.clear();
lcd.print("Vote Recorded");
delay(2000);
lcd.clear();
}
}

// Fingerprint function
int getFingerprintID() {
    uint8_t p = finger.getImage();
    if (p != FINGERPRINT_OK) return -1;

    p = finger.image2Tz();
    if (p != FINGERPRINT_OK) return -1;

    p = finger.fingerSearch();
    if (p != FINGERPRINT_OK) return -1;

    return finger.fingerID;
}
```